Research Project Topic Proposal

Learning a Safety-Aware CNN Cost Map for A*-Based Path Planning in Simulated Autonomous Driving Scenarios

Outline According to the Research Project Rubric

This project studies a hybrid learning-and-search framework for autonomous-driving path planning. The paper will first introduce the motivation for combining classical A* search with a learned cost map. Then, related work will be reviewed, including classical A* search, Neural A* Search, and cost-function learning for autonomous driving. The methodology section will describe the generation of simplified driving-like grid maps, the construction of expert paths, the CNN/U-Net cost-map prediction model, and the final A*-based planner. The experiment section will compare the proposed method with traditional A* baselines using path quality, safety, and efficiency metrics. Finally, the discussion will analyze whether a learned cost map improves planning behavior and what limitations remain for real autonomous driving applications.

Problem, Candidate Solution, and Research Hypotheses

Path planning is a fundamental problem in autonomous driving. Classical A* search can find a valid low-cost path on a graph or grid map, but its performance depends heavily on the cost map and heuristic design. In simple grid planning, the cost is often based only on geometric distance and obstacles. However, autonomous driving requires more than finding the shortest collision-free path. A good driving path should also keep a safe distance from obstacles, remain inside the drivable area, prefer lane-centered motion, and avoid unnecessarily sharp turns.

The candidate solution is to train a convolutional neural network to predict a safety-aware cost map or path-probability map from a simplified bird's-eye-view driving scene. The input will include obstacle locations, road boundaries, lane-center information, start point, and goal point. The learned map will then be combined with a rule-based safety prior and used by a standard A* planner. In this way, the neural network does not directly replace the planner; instead, it provides a learned cost representation that guides an interpretable search algorithm.

The main hypotheses are:

- **H1:** A CNN-predicted cost map can guide A* to generate paths closer to expert safety-aware paths than standard A* using only geometric cost.
- **H2:** The proposed CNN cost map + A* method can improve safety-related metrics, such as obstacle clearance and lane-boundary compliance, compared with traditional A*.
- **H3:** The proposed method can maintain a comparable success rate and path length while improving the quality of planned paths in cluttered driving-like scenarios.

Applied Machine Learning Method and Dataset

The applied machine learning method will be supervised learning using PyTorch. A lightweight U-Net or CNN encoder-decoder will be trained to predict a path-probability map or residual cost map. The training labels will be generated by an expert planner, such as safety-aware A* or Dijkstra search, on synthetic driving-like grid maps. Each map will contain a simplified road area, lane centerline, static obstacles, start location, and goal location. The dataset will be automatically generated and split into training, validation, and test sets. Additional test scenarios with different obstacle densities will be used to evaluate generalization.

The final planning cost may be formulated as a hybrid cost: [$C_{\text{total}} = C_{\text{base}} + \lambda C_{\text{learned}},$]$where$$C_{\text{base}}$ encodes rule-based safety constraints such as obstacles and road boundaries, and C_{learned} represents the CNN-predicted driving preference. The proposed method will be compared with standard A*, Dijkstra, and safety-aware rule-based A*. Evaluation metrics will include success rate, collision rate, path length ratio, planning time, expanded nodes, minimum obstacle clearance, lane deviation, and path smoothness.

References

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- Yonetani, R., Taniai, T., Barekatain, M., Nishimura, M., Kanezaki, A. (2021). *Path Planning using Neural A* Search*. International Conference on Machine Learning.
- Wulfmeier, M., Rao, D., Wang, D. Z., Ondruska, P., Posner, I. (2017). *Large-scale cost function learning for path planning using deep inverse reinforcement learning*. The International Journal of Robotics Research, 36(10), 1073–1087.